INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

TRIPOD, SURVEYING: EXTENSION-LEG TYPE

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies. The original copy of this CID replaces MIL-T-14091E.

1. SCOPE.

1.1 Abstract. This CID covers an extension-leg type tripod to mount and support surveying instruments for use.

2. SALIENT CHARACTERISTICS.

2.1 Description. The tripod shall be a smooth head, extension-leg type equipped with a headcap, threaded instrument fastener, plumb line holding device, plumb bob assembly with case, and carrying sling.

2.1.1 Weight. The weight of the tripod, excluding the plumb bob, case, and carrying sling shall not exceed 15.5 pounds.

2.1.2 Stability. The tripod, when positioned for use, shall be able to support loads of up to 100 pounds without slippage, damage, or permanent deformation of any component.

2.2 Materials. The tripod shall be fabricated from compatible materials, inherently corrosion resistant, or treated to provide protection against corrosion and deterioration. All metal components of the tripod shall be nonmagnetic.

2.2.1 Treatment and painting. Unless otherwise specified (see 6.2), the tripod shall be treated and painted in accordance with the manufacturer's normal method for retail distribution.

Comments, suggestions, or questions on this document should be addressed to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN: CEERD-TS-T, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864 or emailed to pd-te@tec.army.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil .

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2.3 Tripod head. The bearing surface of the tripod head (see firure 1) shall be machined finished with center hole 2.25 to 2.50 inches in diameter to accommodate the instrument fastener and the plumb line holding device. The instrument fastener and the plumb line holding device shall permit the fastening or removal of an instrument by an operator wearing arctic gloves. The tripod head shall be protected with a removable headcap, secured to the tripod head. The head shall be provided with three coplanar surfaces for attachment of the tripod legs.

2.3.1 Instrument fastener. An instrument fastener (see figure 1) shall hold the surveying instrument in place on top of the tripod head. For interface with current equipment the fastener shall be attached to the tripod head by a sliding bar arrangement. The sliding bar arrangement shall be fastened at one end to the bottom of the tripod head to pivot and allow the instrument to be positioned anywhere within head center hole. Threads on the instrument fastener shall be 0.625-11 UNC-2A conforming to ASME B1.1. When the instrument fastener is pushed upward to its extremity, it shall protrude above the tripod head not less than 0.38 inches and not greater than 0.50 inches. When not in use, the instrument fastener shall drop below the surface of the head.

2.3.2 Plumb line holding device. The plumb line holder shall be an industry standard, quick disconnect type allowing the plumb bob to be removed without tools (see figure 1). The plumb line holding device shall suspend a plumb bob from the bottom, center of the instrument fastener. With an instrument fastened to the tripod, the plumb line holding device shall ensure that the pivot point of the plumb bob line will be within 0.50 ±0.06 inches of the top of the tripod head. The total unobstructed plumb line swing shall be not less than ± 5 degrees from the vertical.

2.4 Tripod legs. The tripod legs shall be of the extension-leg type, hinged to the tripod head and extending to an end shoe. The extended length, measured from the center of the hinge to the tip of the end shoe, shall be 70.00 ±2.00 inches and the collapsed length shall be 44.00 ±2.00 inches. The legs shall be adjusted to the desired length with no binding or distortion. A locking device shall secure each leg at any extended position and when fully retracted. An integrated device shall be provided to secure the legs together for transporting the tripod. A device shall be provided for attaching the removed headcap in such a way as not to interfere with the tripod head or equipment operation. A device shall be provided for attaching a standard survey instrument portable battery box.

2.4.1 Hinged attachment. The hinged attachment shall provide a firm connection between the legs and the tripod head. The hinged attachment shall allow the legs to move freely through a minimum 90° arc from vertical to horizontal, and lock in any position within the arc.

2.4.2 End shoes. Each end shoe shall be 4.00 to 7.00 inches long and tapered to a hardened, replaceable point. Use of magnetic material shall be acceptable. Each end shoe shall have a permanently attached foot spur to assist in firmly placing the shoe into ground. The foot spur shall be on the outside of the shoe and extend from the shoe 1.50 ±.06 inches. The end shoe points shall be replaceable without the use special tools.

2.5 Plumb bob assembly. The plumb bob shall have hardened, replaceable points. The points shall be replaceable without the use of special tools. The plumb bob shall be furnished with a minimum of 7.50 feet of non-binding cord, a cord length adjuster, and a plumb line holding device.

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2.5.1 Case. A case shall be provided to hold the plumb bob assembly, additional points, and accessories. The case shall be attached to the tripod and shall not hinder the full movement and operation of the tripod or survey instrument.

2.6 Carrying sling. A carrying sling shall be provided for carrying the tripod with or without an instrument. It shall be padded and adjustable to provide a sufficient length for carrying the tripod over the shoulder.

2.7 Identification marking. Each tripod shall have a durable data plate identifying, as a minimum, the manufacturer's name and part number. The data plate shall be located in a place where the data will be noticeably visible and legible, but not interfere with the operation and maintenance of the tripod.

3. REGULATORY REQUIREMENTS.

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Market acceptability. The tripod offered shall be the latest model of the standard product of the supplier and that has been used for at least one year by commercial organizations.

4.2 Conformance statement. The tripod shall meet the salient characteristics of this CID, conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.3 Examinations. Each tripod shall be examined for compliance with the requirements specified in 2.1 through 2.7. Any modification necessary following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of the performance and dimensional requirements. Non-compliance with any specified requirements shall constitute cause for rejection.

5. PACKAGING.

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Intended use. The tripod covered by this specification is intended for mounting theodolites, levels, electronic distance measuring devices, and other surveying instruments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this CID.
b. If special treatment or color is required (see 2.2.1).
c. Packaging (see 5.1).

6.3 Data requirements. The contracting officer should include in the contract or order, any data requirements, such as maintenance and operation manuals, instructional materials, and technical publications to be furnished with each tripod.

6.4 Provisioning. The contracting officer should include in the contract or order, provisioning requirements for repair parts, maintenance tools, as necessary; and instructions regarding shipment of the tripod.

6.5 National Stock Numbers (NSN's). The following NSN is assigned to a product that conforms to this CID: 6675-00-641-3572. This may not be indicative of all possible NSNs associated with the CID.

6.6 Subject term (key words) listing.

Mount
Stand
Support
Rig
Pedestal
Base
Instrument

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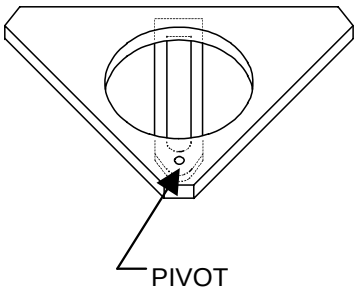

TRIPOD HEAD
SHOWING CENTER HOLE
AND ATTACHMENT BAR.

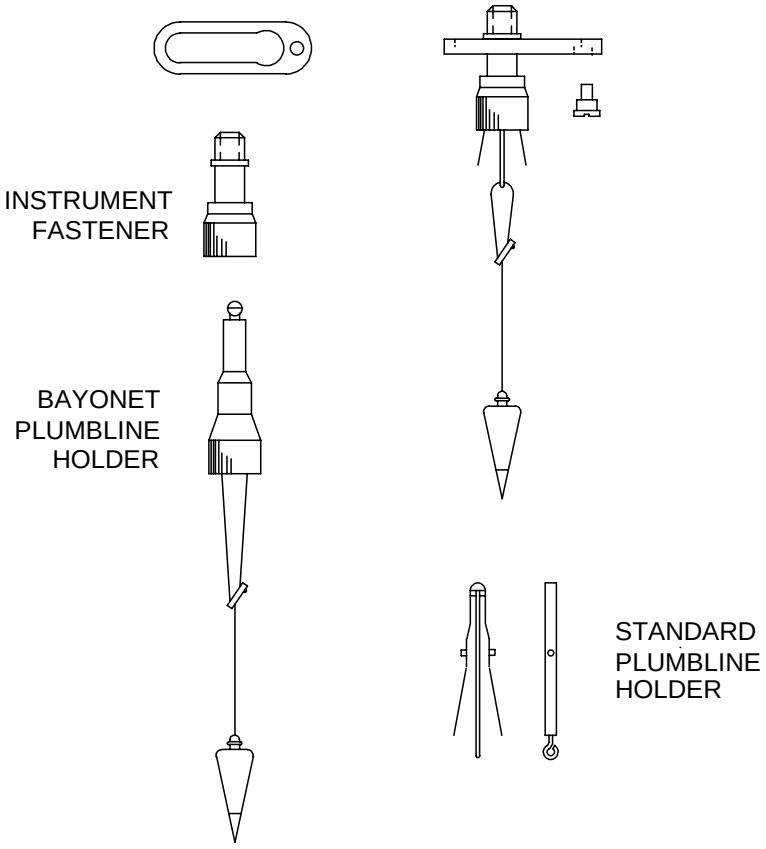

FIGURE 1. Instrument fasteners with plumb line holders - Example.

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